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Bi_2Te_3 nanowire array preparation and surface polishing by electrochemical deposition

D. Ren*[1], T. Guo[1], Z. Yang[1] and J. Mu[2]

The quantum confinement effect and phonon scattering in low dimensional nanostructure is effective in the improvement of the efficiency of thermoelectric generators. Bi_2Te_3 nanowire arrays were prepared in Al_2O_3 templates through the use of electrochemical deposition. Chemical mechanical planarisation was employed to level the upper and lower surfaces of Bi_2Te_3 samples. The XRD (X-ray Diffractometer) test showed that the crystal grain of Bi_2Te_3 nanowire was 20–30 nm, attaining a size where the structural properties of the structure induced the quantum confinement effect. It was also indicated that there were dense structures in the fabricated nanowires, the nanowire ends kept at the same level as the hole's edge, solving the problem for the following assembly of thermoelectric transducers. Based on the procedures presented in this paper, there is promise in the future enhancement of the Seebeck coefficient of this nanowire array by introducing nanodoping to the structure.

Keywords: Thermoelectric conversion, Bi_2Te_3 nanowire, Electrochemical deposition, Porous AAO template, Chemical mechanical planarisation

Introduction

About one-third of the heat in the modern industry production is emitted in the form of waste heat. With the increasing threats of environmental destruction and energy crises, green energy and relevant components have begun to command more attention. By integrating heat dissipation and power generation, thermoelectric generators have become research hotspots in recent years for their unique advantages: no moving components, no noise, little environmental impact and high reliability. However, their thermoelectric conversion efficiency is still inferior to traditional materials at present, making them unsuitable for wide applications. Fortunately, the development of novel materials and nanotechnology provides new approaches to solve this problem. With the efficiency barrier removed, potential applications are expanded to integrated circuits, electronic components, biomedical engineering and aerospace technologies.[1]

There are two approaches to improve the thermoelectric figure-of-merit parameter, ZT: enhancement of the electrical conductivity and reduction of the thermal conductivity. The low-dimensional treatment of nanostructures can balance the control for both. When the physical dimensions of Bi–Te materials are reduced to the average free path of the carriers (8–14 nm), phonon scattering is enhanced while the heat conductivity is restrained. Meanwhile, the electronic state density is raised to near Fermi levels, improving the Seebeck coefficient and electrical conductivity. Low-dimensional methods include the nanowire array, nanofilm, superlattice, quantum dot and so on. Compared with other nanostructures, the nanowire array is more promising as a result of its good thermoelectric characteristics and fabrication convenience. Electrochemical deposition has unique advantages when fabricating nanowires in templates. In addition to the introduction of quantum confinement effect and phonon scattering as a result of low dimensionality, ternary or quaternary solution deposition can also be applied to facilitate doping with doping of elements such as Sb and Se, further improving thermoelectric performance.[2]

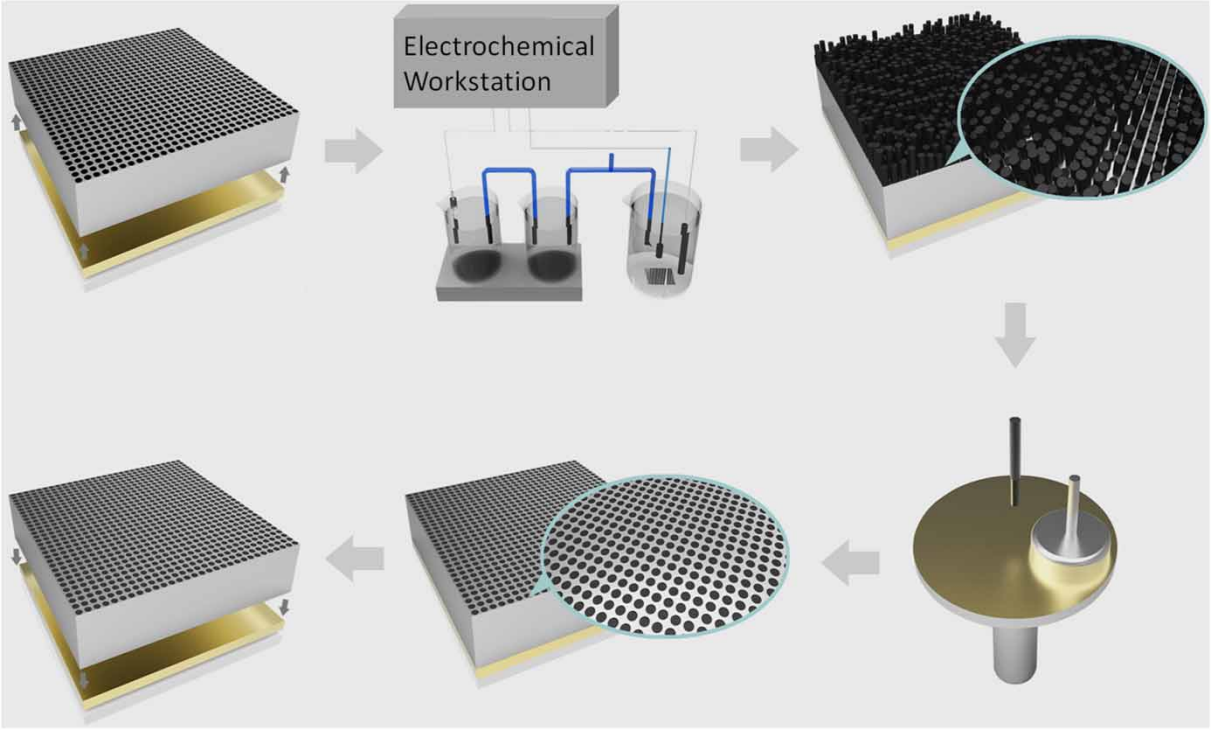
Currently, there are primarily six types of thermoelectric materials with superior performance. The comprehensive operating temperature for these materials spans from 200 to 1000 K. Bi–Te compounds are mainly applicable for temperatures of 200–400 K,[3,4] Pb–Te are 600–900 K,[5,6] $AgSbTe_2$ compounds are 500–800 K,[7,8] skutterudite materials are 800–900 K,[9,10] skutterudite compounds are 800–900 K[9,10] and Bi–Co oxide thermoelectric materials are mostly 900–1000 K.[11,12]

Bi–Te compounds have optimal thermoelectric performance under room temperature (200–400 K). Marcus Scheele fabricated nanoparticles with a grain size of 10 nm or less by discharge plasma sintering, and significantly reduced the thermal conductivity, thus raising the thermoelectric power factor.[13] The type-p material with a maximum ZT value under room temperature is $Bi_{0.5}Sb_{1.5}Te_3$, a super-lattice film fabricated by Venkatasubramanian with a ZT value of 2·4 at 300 K.[14]

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1 Procedure of Bi_2Te_3 nanowire array preparation

According to the differences in principles and structures, thermoelectric materials of Bi_2Te_3 with low dimensionality can be divided into three categories, that is, superlattice, film and nanowire, for which, research reports on Bi_2Te_3 nanowires account for approximately 12%. Current methods for the preparation of Bi_2Te_3 nanowires mainly focus on the solvothermal synthesis of nanoparticles and template-based electrochemical deposition. The minimum diameter of Bi_2Te_3 grains and nanowires made from solvothermal synthesis can potentially reach the submicron level. In comparison, grains obtained from electrochemical deposition exhibit diameters that could be as small as 20 nm. Additionally, the deposition process does not require high temperatures and high pressures, making it a low-cost and energy- efficient solution.

In the present reports of Bi_2Te_3 nanowire growth, materials are usually loaded into the nanopores using a vacuumising process that is difficult to execute. We present an easy and non-hazardous approach for the Bi_2Te_3 nanowire growth, which also offers an effective method for nanowire preparation.

Using electrochemical deposition, we prepared Bi_2Te_3 nanowire arrays in Al_2O_3 templates, having respective hole diameters of 25, 70 and 100 nm. The XRD (X-ray Diffractometer) analysis indicated that the crystal grain of the nanowires obtained was 23 nm, reaching a scale where quantum confinement occurs.

Experimental

Experiment instruments

The field-emission SEM was performed using the FEI Quanta 200 FEG and its EDS (Energy Dispersive Spectrometer) and EBSD (Electron Backscattered Diffraction) integrated analysis accessories (American FEI Company), generally in high-vacuum mode, 1·2 nm (30 kV) and 3·0 nm (1 kV). In addition to facilitating SEM imaging, metal spraying was also applied to the surfaces of the AAO (Anodic Aluminum Oxide)-nanowire samples for 30 s to improve electrical conductivity. An X-ray diffractometer was employed using Rigaku's D/max2250HB + /PC to analyse the sample phase structure with scanning angle of 10°–80°. Chemical–mechanical planarisation was conducted with a TegraForce-35 precision polishing machine (Danish Struers). A cerium oxide solution (Beijing Boyu Gaoke New Material Technology Co., Ltd.) was applied as a polishing fluid while the primary grain size of the cerium oxide was 25 nm.

Chemical agents

The electrolyte for the Bi_2Te_3-based nanowires was a HNO_3 solution, which was composed of $Bi(NO_3)_3{\cdot}5H_2O$ and TeO_2. The concentrations for the electrolytes are given in Table 1. Because the surface tension of water molecules prevents liquids from entering nanoholes, we added the active agent $C_{12}H_{25}SO_4$ to facilitate liquid entry into the 25 nm template holes. The solution was incubated in the dark for 24 h after preparation until all solute granules were fully dissolved, forming a clear, transparent solution.

Experimental process

First, a layer of metallic electrode of a 200 nm thickness was sputtered at the bottom of the Al_2O_3 template with

Table 1 Concentrations of components in the electrolyte solution

$[Bi^{3+}]$ (mmol L^{-1})	$[HTeO_2^+]$ (mmol L^{-1})	$[HNO_3]$ (mmol L^{-1})	$[Na\ C_{12}H_{25}SO_4]$ (mmol L^{-1})
7	10	1	0·15

dimensions of 32×32 mm and thickness of 100 nm. In the three-electrode circuit, a saturated calomel electrode was used as the reference electrode, and a titanium electrode was used as the auxiliary electrode, which had an area much larger than 20 cm^2. After connecting the components to the three-electrode reaction tank, the reaction was carried out for 4 h under the deposition voltage of −0·068 V. After that, the solution was poured out, and the template was dried. Following the dry step, the template was treated with a precision polisher to get a smooth upper surface. Finally, the gold layer on the lower surface was removed via the etching method. These processes conveniently prepare the template for the nanowire array assembly in the next step (Fig. 1).

Results and discussions

SEM and EDS analysis

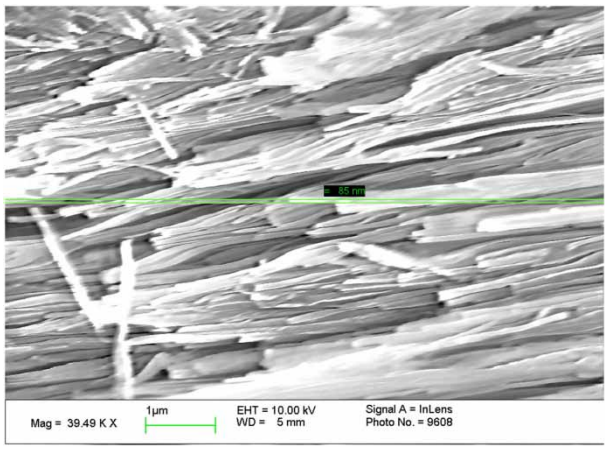
The addition of chemically and mechanically active agents yielded Bi_2Te_3 nanowires with a 25 nm diameter, improving upon previous results. By properly controlling the concentration (10 mmol L^{-1} of Bi^{3+} ions and 7 mm mol L^{-1} of TeO^{2-} ions) and the sedimentation potential, a stable Bi_2Te_3 phase compound was obtained, as shown in Fig. 2 and Fig. 3, meeting the requirement of further integration of thermoelectric generators (Table 2).

The comparison between the standard XRD spectrum of Bi_2Te_3 and the detected one are shown above (Fig. 4). We calculated the crystal grain size using the Scherrer formula: $D = k\lambda/\beta\cos\theta$, where $K = 1$ and the radiation wavelength is $\lambda = 0{\cdot}154056$ nm. Selecting $2\theta = 27{\cdot}66°$, FWHM (Full Width at Half Maximum) $B = 0{\cdot}57°$ at $2\theta = 27{\cdot}66°$ and the instrument width $b = 0{\cdot}167°$, the average size of the Bi_2Te_3 crystal grain was calculated, where $D = 22{\cdot}15$ nm. The results are corroborated by the direct SEM observations.

Morphology after CMP (Chemical Mechanical Polishing)

Repeating the experiment and respectively reducing metal spraying time, the crystal particles looked denser when viewed via SEM; the size of the metal particles was reduced significantly as well.

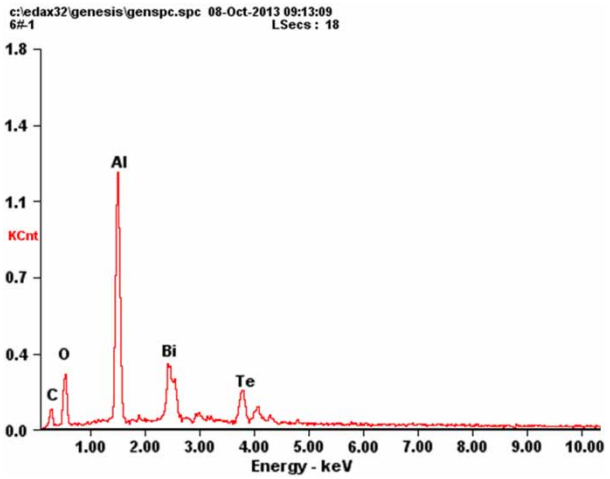
The crystal structure of the Bi_2Te_3 nanowire arrays was determined by conditions of the electrochemical deposition, such as the pH of the solution, the ion concentration and the deposition voltage. Since the diameters of the nanopores of the porous anodic alumina template were not absolutely consistent, and the electric field in the solution was not precisely in a parallel distribution, the growth speeds of Bi_2Te_3 nanowires were not absolutely identical. Because of the variation in growth rates, there were inconsistent heights in the generated nanowire.



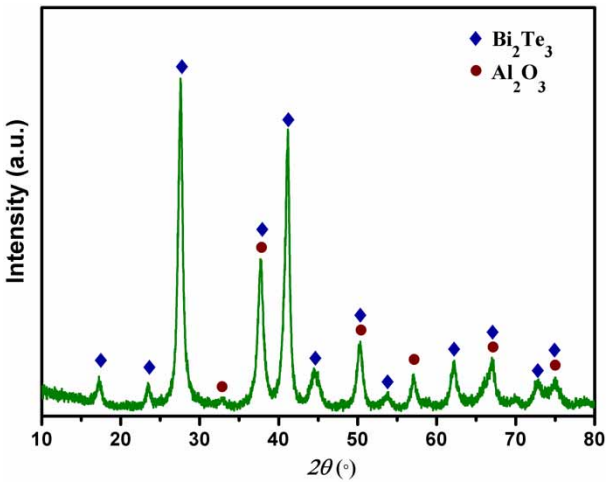
2 Bi_2Te_3 nanowires observed by SEM

Table 2 The ratio of the atom number EDS analysis

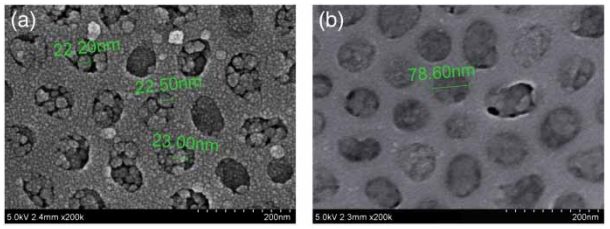
Element	Wt%	At.%
Bi	53·66	42·42
Te	46·34	58·58



3 Bi_2Te_3 nanowires analysed by EDS



4 XRD analytical consequence



5 Morphology after CMP

However, with CMP processing, the templates and nanowire arrays were planarised, creating structures of the same height, which could significantly reduce the parasitic resistance in the following electrical connection. Parasitic resistance is the biggest obstacle in augmentation of thermoelectric power output.

Though promoting conductivity in a Bi_2Te_3 matrix, the conductivity of Al_2O_3 is very weak. To get a clear SEM image, it is necessary to spray gold on the surface of the sample. The left and the right pictures in Fig. 5 are the results after spraying gold for 50 and 15 s, respectively. It can be seen that, in the left picture, the size of the filler particles in the nanopore, Bi_2Te_3 crystal grains, are 1/2–1/3 of the diameter of the nanopore. These grains and surface of the template are coated with a tiny moss-like substance, which is the sprayed gold. In the picture on the right, because the spraying time was markedly reduced, there are only some tiny gold particles covering the surface of the sample.

Conclusion

Bi_2Te_3 nanowire arrays were prepared in AAO templates by using electrochemical deposition. Chemical–mechanical planarisation was employed to produce a flat upper surface for the following electrode integration. As demonstrated, the nanodoping process coupled with the control and coordination of factors such as electron density distribution, selective scattering of electrons, valence band/conduction band and external carrier concentration, promotes enhancement of the Seebeck coefficient of nanowire arrays, expanding the platform of nanostructure-based thermoelectric generators.

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